

Aaryan Kainth

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WORK EXPERIENCE

Cadence (cadence.com)

Goleta, CA

R&D Design Engineering Intern

September 2025 – February 2026

- **Designed and built a shoe test setup** to characterize transition force from low- to high-friction surfaces
- Wrote an SOP for the test setup, documenting how to make important components and writing detailed testing procedures
- Researched, prototyped, and tested various fall-prevention shoe designs on the way to production

Walt Disney Imagineering R&D (sites.disney.com/waltdisneyimagineering)

Glendale, CA

Lab Associate

June 2025 – September 2025

- **Designed and fabricated an integrated remote control box** to control a next-generation aquatic performance robot using SolidWorks, laser cutting, FDM 3D printing, and sheet metal bending
- Designed and constructed parts, tooling, and testing equipment for said robot, including mounts, adapters, a boat, and a lift to get the robot in and out of the water; utilized speed rail, waterjet cutting, and SLS 3D printing
- **Designed and constructed an electronic, lightsaber-themed ping-pong tournament trophy within 24 hours**
- Created a SolidWorks Hole Wizard preset to automatically create parameter-driven holes for heat set inserts for 3D prints

Hawkes Lab (hawkeslab.com)

Goleta, CA

Undergraduate Research Assistant

May 2023 – June 2025

- **First authored a paper** detailing the design and fabrication of a mounted anchor launcher that allows extraterrestrial rovers to traverse soft-sand slopes; **presented it at the International Conference for Robotics and Automation in 2025**
- Designed the clutch mechanism used in the anchor launcher
- Learned effective experimental design to create a model for anchor launch distance using Excel and MATLAB

Plume Design, Inc. (plume.com)

Palo Alto, CA

Hardware Team Summer Intern

June 2021 – August 2021

- Wrote Python testing scripts to quantify performance of 3 different kinds of product antennas
- Engineered a basic dipole antenna for 5 GHz bandwidth
- Learned how to use iBwave antenna simulation software, and trained additional team members on how to use it
- **Leveraged Python and Raspberry Pi to collect and interpret data** for wall mounted fall-detection prototype

PROJECTS

Capstone Project - *Pinch-Free Animatronic Shells*

September 2025 – Present

- **Connected relevant stakeholders at WDI R&D and UCSB's ME Capstone team** to initiate capstone project
- Designed thematic shells to cover hazardous pinch points on robotic arms, including shoulder, elbow, and wrist joints
- **Leveraged ROS2 and Dynamixel MoveIt** to animate an OpenManipulator-X arm to use as a testbed

Junior Design Project - *CD Jukebox*

April 2025 – June 2025

- Designed a belt-driven lead screw system to adjust the linear position of a servo arm
- Wrote and tuned a PD controller in C++ to precisely drive the arm to within 2mm of target

Eagle Project - *Wheelchair Accessible Animal Tables*

February 2022 – June 2022

- Led 9 volunteers over the course of 103 man hours to build 3 ADA-compliant accessible animal tables for a local petting zoo, allowing them to better serve visitors who use wheelchairs

EDUCATION

University of California, Santa Barbara

Goleta, CA

Bachelor of Science (B.S.) in Mechanical Engineering: 3.96 Cumulative GPA

Expected June 2026

Master of Science (M.S.) in Mechanical Engineering: Beginning June 2026

Expected June 2027

SKILLS

Programming Languages: Python, MATLAB, C++

Manufacturing Skills: CAD, CNC machining, laser/waterjet cutting, design for manufacturing, rapid prototyping

Basic Proficiencies: Microsoft Office, Google Suite, LaTeX/Overleaf